8 **(a)** State the experimental evidence that suggests that the process of radioactive decay is spontaneous.

.....

.....

..... [1]

(b) The radioactive isotope iodine-131 ($^{131}_{53}\mathrm{I}$) undergoes beta-decay. It has a decay constant of 8.6×10^{-2} day^{-1}.

In nuclear medicine, iodine-131 is injected into or swallowed by a patient to target and destroy cancerous cells in the thyroid gland.

(i) Suggest why alpha radiation is not suitable for destroying targeted cancerous thyroid cells.

.....

.....

..... [1]

(ii) A sample of blood contains 1.2×10^{-9} g of iodine-131.

Determine, for this sample of blood,

1. the activity of the iodine-131,

activity = ... Bq [3]

2. the time for the activity of the iodine-131 to be reduced to 1/50 of the activity calculated in **(ii)1**.

time = ... s [2]

[Total: 7]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN